**7** Some people have blood type A and some people have brown hair. A person may have both, just one, or neither of these characteristics.

A person is selected at random.

A is the event 'the person has blood type A'.
B is the event 'the person has brown hair'.

You are given the following probabilities.

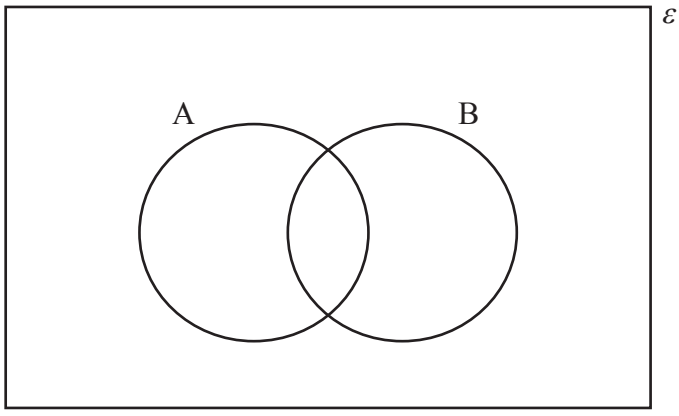
$P(A \cap B') = 0.3382$, $P(A' \cap B) = 0.0682$ and $P((A \cup B)') = 0.5518$.

**(a)** Show this information on the Venn diagram in the Printed Answer Booklet. **[2]**

**(b)** Determine whether A and B are independent events. **[4]**

**(c)** Explain whether A and B are mutually exclusive events. **[1]**

**7(a)**




**7(b)**

**7(c)**